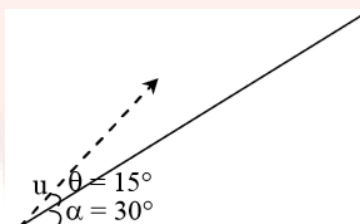


JEE 2019

1. A plane is inclined at an angle $\alpha = 30^\circ$ with respect to the horizontal. A particle is projected with a speed $u = 2 \text{ ms}^{-1}$, from the base of the plane, making an angle $\theta = 15^\circ$ with respect to the plane as shown in the figure. The distance from the base, at which the particle hits the plane is close to:

(Take $g = 10 \text{ ms}^{-2}$)



A. 26 cm

B. 14 cm

C. 20 cm

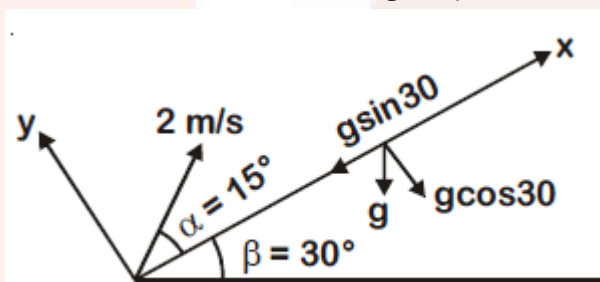
D. 18 cm

Ans. C

Sol:

As we know that

$$\text{Time of flight} = \frac{2u \sin \alpha}{g \cos \beta}$$



Range

$$T = \frac{u \sin 15^\circ}{g \cos 30^\circ} = \frac{2 \times 2 \sin 15^\circ}{10 \cos 30^\circ} = \left(\frac{4 \sin 15^\circ}{10 \cos 30^\circ} \right)$$

$$S = t(2 \cos 15^\circ) - \frac{g \sin 30^\circ t^2}{2} = \frac{4 \sin 15^\circ}{10 \cos 30^\circ} (2 \cos 15^\circ) - \frac{10 \sin 30^\circ}{2} \left(\frac{4 \sin 15^\circ}{10 \cos 30^\circ} \right)^2$$

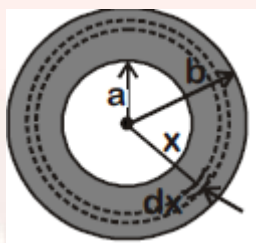
$$S = 20 \text{ cm}$$

2. Space between two concentric conducting spheres of radii a and b ($b > a$) is filled with a medium of resistivity ρ . the resistance between the two spheres will be:

- A. $\frac{\rho}{4\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$ B. $\frac{\rho}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$
 C. $\frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$ D. $\frac{\rho}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$

Ans. D

Sol: Resistance of spherical shell is



$$dR = \frac{\rho dx}{4\pi x^2}$$

$$R = \int dR = \frac{\rho}{4\pi} \int_a^b \frac{dx}{x^2}$$

$$R = \frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$$

3. In the formula $X = 5YZ^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units ?

- A. $[M^{-2} L^0 T^{-4} A^{-2}]$ B. $[M^{-2} L^{-2} T^6 A^3]$
 C. $[M^{-3} L^{-2} T^8 A^4]$ D. $[M^{-1} L^{-2} T^4 A^2]$

Ans. D

Sol: Given that

$$X = 5Y Z^2$$

$$X = \text{capacitance} = \frac{Q}{V}$$

$$= \frac{Q^2}{\omega} = \frac{Q^2}{F.L}$$

$$= \frac{A^2 L^2}{M^1 L^1 T^{-2} \times L}$$

$$= A^2 M^{-1} L^{-2} T^4$$

$$= M^{-1} L^{-2} T^4 A^2$$

$$Z = B = \frac{F}{i \ell} = \frac{M^1 L^1 T^{-2}}{A^1 L^1}$$

$$= M^1 L^0 T^{-2} A^{-1}$$

$$Z^2 = M^2 T^{-4} A^{-2}$$

$$Y = \frac{X}{Z^2} = \frac{M^{-1} L^{-2} T^4 A^2}{M^2 T^{-4} A^{-2}}$$

$$= M^{-3} L^{-2} T^8 A^4$$

4. A spaceship orbits around a planet at a height of 20 km from its surface. Assuming that only gravitational field of the planet acts on the spaceship, what will be the number of complete revolutions made by the spaceship in 24 hours around the planet?

[Given : Mass of planet = 8×10^{22} kg, Radius of planet = 2×10^6 m,

Gravitational constant $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$]

- | | |
|-------|-------|
| A. 17 | B. 13 |
| C. 11 | D. 9 |

Ans. C

Sol:

Orbital speed of spaceship is

$$v = \sqrt{\frac{GM}{(R + h)}} \quad \dots(i)$$

Here, $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$,

$$M = 8 \times 10^{22} \text{ kg}$$

$$R + h = (2 \times 10^6 + 20 \times 10^3) \text{ m}$$

$$= 2.02 \times 10^6 \text{ m}$$

So, substituting these values in Eq. (i), we get

$$v = \sqrt{\frac{6.67 \times 10^{-11} \times 8 \times 10^{22}}{2.02 \times 10^6}}$$

$$= 16 \times 10^3 \text{ ms}^{-1}$$

Time period of rotation of spaceship will be

$$T = \frac{2\pi(R + h)}{v}$$

$$\Rightarrow T = \frac{2\pi \times 2.02 \times 10^6}{1.6 \times 10^3}$$

$$\approx 8 \times 10^3 \text{ s} = \frac{8 \times 10^3}{60 \times 60} (\text{h})$$

$$= 2.2 \text{ h}$$

So, number of revolution made by spaceship in 24h,

$$n = \frac{24}{T} = \frac{24}{2.2} \approx 11 \text{ rev}$$

5. A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m . If this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be:

A. $\frac{2m}{\pi}$

B. $\frac{4m}{\pi}$

C. $\frac{3m}{\pi}$

D. $\frac{m}{\pi}$

Ans. B

Sol:

Wire length will be same in both area,

$$\Rightarrow 4a = 2\pi r \Rightarrow r = \frac{4a}{2\pi} = \frac{2a}{\pi}$$

Hence, area of circular loop formed is, $A' = \pi r^2$

$$= \pi \left(\frac{2a}{\pi} \right)^2 = \frac{4a^2}{\pi}$$

Magnitude of magnetic dipole moment of circular loop will be

$$m' = IA' = I \frac{4a^2}{\pi}$$

Ratio of magnetic dipole moment of both shapes is

$$\frac{m'}{m} = \frac{I \cdot \frac{4a^2}{\pi}}{Ia^2} = \frac{4}{\pi} \Rightarrow m' = \frac{4m}{\pi} (A - m)$$

6. A coil of self-inductance 10 mH and resistance 0.1Ω is connected through a switch to a battery of internal resistance 0.9Ω . After the switch is closed, the time taken for the current to attain 80% of the saturation value is : [take $\ln 5 = 1.6$]

- A. 0.016 s B. 0.324 s
C. 0.103 s D. 0.002 s

Ans. A

Sol:

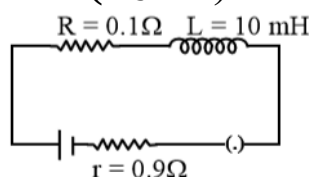
In given circuit

Inductance of circuit is

$$L = 10 \text{ mH} = 10 \times 10^{-3} \text{ H}$$

Resistance of circuit is

$$R = (R_s + r) = 0.1 + 0.9 = 1\Omega$$



$$\text{Given, } i = \frac{80i_0}{100} = 0.8i_0$$

$$i = i_0 \left[1 - e^{-\frac{Rt}{L}} \right]$$

$$0.8i_0 = i_0 \left[1 - e^{-\frac{Rt}{L}} \right]$$

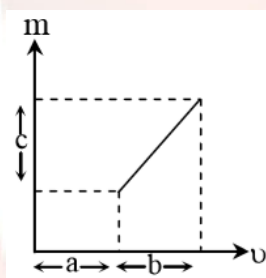
$$e^{-\frac{Rt}{L}} = 0.2 = \frac{1}{5}$$

$$-\frac{Rt}{L} = \ln(5)$$

$$t = \frac{L}{R} \ln(5)$$

$$= 0.016 \text{ sec}$$

7. The graph shows how the magnification m produced by a thin lens varies with image distance v . What is the focal length of the lens used?



A. $\frac{b^2}{ac}$
C. $\frac{b^2c}{a}$

B. $\frac{a}{c}$
D. $\frac{b}{c}$

Ans. D

Sol:

As we know that

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$1 - m = \frac{v}{f}$$

$$m = 1 - \frac{v}{f}$$

$$v = a, m = 1 - \frac{a}{f}$$

$$\text{at } v = a + b, m = 1 - \frac{a + b}{f}$$

$$m_2 - m_1 = c = \frac{b}{f}$$

$$\therefore f = \frac{b}{c}$$

8. In Li^{++} , electron in first Bohr orbit is excited to a level by a radiation of wave length λ . When the ion gets de excited to the ground state in all possible ways (including intermediate emissions), a total of six spectral lines are observed. What is the value of λ ? (Given: $h = 6.63 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ ms}^{-1}$)

- A. 10.8 nm B. 11.4 nm
C. 9.4 nm D. 12.3 nm

Ans. A

Sol:

Energy observed by electron from incident photon of wavelength λ is

$$\Delta E = \frac{hc}{\lambda} = 13.6 \times 9 - 0.85 \times 9$$

$$\lambda = \frac{hc}{9(13.6 - 0.85)}$$

$$\lambda = \frac{1237}{9 \times 12.75} \text{ nm}$$

$$\approx 10.8 \text{ nm}$$

9. A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is converted into a uniform disc of radius $2R$. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the disc about its axis and I_2 be the moment of inertia of the new sphere about its axis. The ratio I_1/I_2 is given by:

- A. 65 B. 285
C. 140 D. 185

Ans. C

Sol:

The moment of inertia of the disc about its axis is given by

$$I_1 = \frac{\left(\frac{7m}{8}\right)(2R^2)}{2} = \frac{7}{4} MR^2$$

the moment of inertia of the disc about its axis

$$I_2 = \frac{2}{5} \left(\frac{M}{8}\right) R_1^2 = \frac{MR_1^2}{80}$$

$$R = 2R_1$$

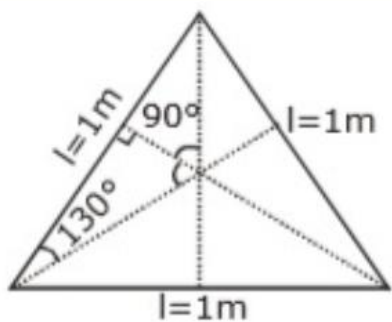
$$\frac{I_1}{I_2} = \frac{7}{4} \times 80 = 140$$

10. The magnitude of the magnetic field at the centre of an equilateral triangular loop of side 1 m which is carrying a current of 10 A is : [Take $\mu_0 = 4\pi \times 10^{-7} \text{NA}^{-2}$]

- A. $3 \mu\text{T}$ B. $1 \mu\text{T}$
C. $9 \mu\text{T}$ D. $18 \mu\text{T}$

Ans. D

Sol:



m due to single wire

$$B = \frac{\mu_0}{4\pi d_1} (\sin 60^\circ + \sin 60^\circ)$$

$$\therefore B_{\text{net}} = 3B_1$$

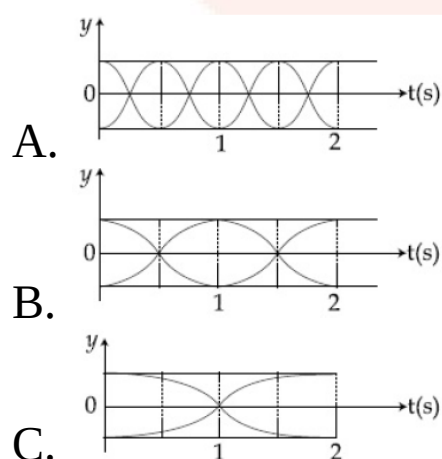
$$\left(\frac{3\mu_0}{4\pi} \frac{I}{2\sqrt{3}} (2\sin 60^\circ) \right)$$

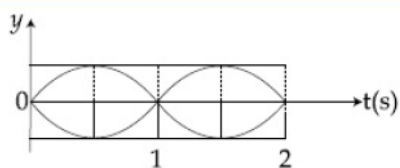
$$B = \frac{\mu_0 i}{4\pi d} (\sin \theta_1 + \sin \theta_2)$$

$$= \frac{\mu_0 i (2\sqrt{3})}{4\pi l} [2\sin 60^\circ]$$

$$= 18\mu T \text{ Am}$$

11. The correct figure that shows, schematically, the wave pattern produced by superposition of two waves of frequencies 9 Hz and 11 Hz, is:





D.

Ans. A

Sol:

Beat frequency is given by, $f_{\text{beat}} = f_1 - f_2$

Here, $f_1 = 11 \text{ Hz}$ and $f_2 = 9 \text{ Hz}$

Now $f_{\text{beat}} = f_1 - f_2 = 11 - 9 = 2 \text{ Hz}$

12. The elastic limit of brass is 379 MPa. What should be the minimum diameter of a brass rod if it is to support a 400 N load without exceeding its elastic limit ?

- A. 0.90 mm B. 1.00 mm
C. 1.16 mm D. 1.36 mm

Ans. C

Sol:

Let d = diameter of brass.

then stress of brass rod is given by

$$\text{Stress} = \frac{F}{A} = \frac{4F}{\pi d^2} \quad \left[\because A = \frac{\pi d^2}{4} \right]$$

$$= \frac{400 \times 4}{\pi d^2} = 379 \times 10^6 \quad [\because F = 400 \text{ N}]$$

$$d = \sqrt{1.34} \times 10^{-3}$$

$$\approx 1.1 \times 10^{-3}$$

13. In free space, a particle A of charge $1 \mu\text{C}$ is held fixed at a point P. Another particle B of the same charge and mass $4 \mu\text{g}$ is kept at a distance of 1 mm from P. If B is released, then its velocity at a distance of 9 mm from P is :

[Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{NM}^2\text{C}^{-2}$]

- A. $3.0 \times 10^4 \text{ m/s}$ B. $1.5 \times 10^2 \text{ m/s}$
C. $2.0 \times 10^3 \text{ m/s}$ D. 1.0 m/s

Ans. C

Sol:

Change in potential energy of system A and B

$$\begin{aligned} &= kq_1q_2 \left[\frac{1}{d_1} - \frac{1}{d_2} \right] \\ &= k \times 10^{-12} \left[1 - \frac{1}{9} \right] \times 10^3 \\ &= 9 \times 10^9 \times 10^{-9} \left[\frac{8}{9} \right] = 8 \end{aligned}$$

By the conservation of energy

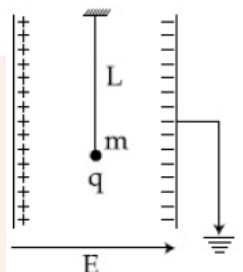
Kinetic energy of charged particle B = change in potential energy of system A and B.

$$\frac{1}{2}mv^2 = 8$$

$$\begin{aligned} v &= \sqrt{\frac{16}{m}} = \sqrt{\frac{16}{4 \times 10^{-9}}} \\ &= 2 \times 10^4 \times \sqrt{10} \\ &\approx 6 \times 10^4 \text{ m/s} \end{aligned}$$

Of mass is considered as $m = 4 \times 10^{-6} \text{ kg}$ then $v = 2 \times 10^3 \text{ m/s}$.

14. A simple pendulum of length L is placed between the plates of a parallel plate capacitor having electric field E , as shown in figure. Its bob has mass m and charge q . The time period of the pendulum is given by:



A. $2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$

B. $2\pi \sqrt{\frac{L}{\sqrt{g^2 - \left(\frac{qE}{m}\right)^2}}}$

C. $2\pi \sqrt{\frac{L}{\sqrt{g^2 - \left(\frac{q^2 E^2}{m^2}\right)}}}$

D. $2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$

Ans. A

Sol:

Net acceleration is, $a_{\text{net}} = \sqrt{a_1^2 + a_2^2}$

$$a_{\text{net}} = \frac{\sqrt{(mg)^2 + (qE)^2}}{m}$$

$$T = 2\pi \sqrt{\frac{L}{a_{\text{net}}}}$$

$$\therefore T = 2\pi \sqrt{\frac{L}{\sqrt{g^2 + \left(\frac{qE}{m}\right)^2}}}$$

15. One mole of an ideal gas passes through a process where pressure and volume obey the relation $P = P_0 \left[1 - \frac{1}{2} \left(\frac{V_0}{V} \right)^2 \right]$. Here P_0 and V_0 are constants. Calculate the change in the temperature of the gas if its volume change from V_0 to $2V_0$.

A. $\frac{1}{2} \frac{P_0 V_0}{R}$

B. $\frac{5}{4} \frac{P_0 V_0}{R}$

C. $\frac{1}{4} \frac{P_0 V_0}{R}$

D. $\frac{3}{4} \frac{P_0 V_0}{R}$

Ans. B

Sol:

Ideal gas equations

$$PV = nRT$$

$$T = \frac{PV}{nR} \quad n = 1 \text{ mole}$$

$$T = \frac{V}{R} \left[P_0 \left[1 - \frac{1}{2} \left(\frac{V_0}{V} \right)^2 \right] \right] \quad \left(\text{Given } P = P_0 \left[1 - \frac{1}{2} \left(\frac{V_0}{V} \right)^2 \right] \right)$$

$$T_i = \frac{V_0}{R} \left[P_0 \left[1 - \frac{1}{2} \left(\frac{V_0}{V_0} \right)^2 \right] \right]$$

$$T_f = \frac{2V_0}{R} \left[P_0 \left[1 - \frac{1}{2} \frac{V_0^2}{(2V_0)^2} \right] \right]$$

$$T_i = \frac{P_0 V_0}{2R} \quad T_f = \frac{7}{54} \frac{P_0 V_0}{R}$$

$$\therefore \Delta T = T_f - T_i$$

$$= \frac{7}{4} \frac{P_0 V_0}{R} - \frac{P_0 V_0}{2R} = \frac{5}{4} \frac{P_0 V_0}{R}$$

16. A source of sound S is moving with a velocity of 50 m/s towards a stationary observer. The observer measures the frequency of the source as 1000 Hz. What will be the apparent frequency of the source when it is moving away from the observer after crossing him? (Take velocity of sound in air is 350 m/s)

- A. 750 Hz B. 857 Hz
C. 807 Hz D. 1143 Hz

Ans. A

Sol:

When source is moving towards stationary observer, frequency observed is more than source frequency due to doppler's effect, it is given by

$$f_{\text{observed}} = f \left(\frac{v}{v - v_s} \right)$$

$$F' = \left(\frac{V}{V - 50} \right) F_{\text{source}}$$

$$\therefore F_{\text{source}} = \frac{1000 \times 300}{350}$$

When source moves away from stationary observer, observed frequency will be lower due to Doppler's effect and it is given by

$$\begin{aligned} f'' &= \left(\frac{V}{V + 50} \right) \cdot F_{\text{source}} \\ &= \frac{300}{350} \times \frac{350}{400} \times 1000 \\ &= 750 \text{ Hz} \end{aligned}$$

17. When heat Q is supplied to a diatomic gas of rigid molecules, at constant volume its temperature increases by ΔT . The heat required to produce the same change in temperature, at a constant pressure is :

- A. $\frac{3}{2}Q$ B. $\frac{7}{5}Q$
C. $\frac{2}{3}Q$ D. $\frac{5}{3}Q$

Ans. B

Sol:

Heat supplied at constant volume is given by

$$Q = mC_v \, dT$$

Heat supplied at constant pressure is given by

$$Q' = nC_p \, dT$$

So we have

$$\frac{Q'}{Q} = \frac{C_p}{C_v} = \gamma = \frac{7}{5} \quad (\text{as given gas is diatomic gas})$$

$$Q' = \frac{7}{5}Q$$

18. In an experiment, brass and steel wires of length 1 m each with areas of cross section 1 mm^2 are used. The wires are connected in series and one end of the combined wire is connected to a rigid support and other end is subjected to elongation. The stress required to produce a net elongation of 0.2 mm is,

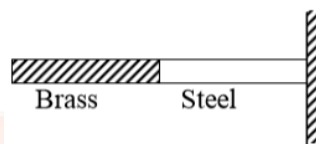
[Given, the Young's Modulus for steel and brass are, respectively, $120 \times 10^9 \text{ N/m}^2$ and $60 \times 10^9 \text{ N/m}^2$]

- A. $1.2 \times 10^6 \text{ N/m}^2$ B. $0.2 \times 10^6 \text{ N/m}^2$
C. $1.8 \times 10^6 \text{ N/m}^2$ D. $4.0 \times 10^6 \text{ N/m}^2$

Ans. Bonus

Sol:

In given experiment, a composite wire is stretched by a force F .



Net elongation in the wire = elongation in brass wire + elongation in steel wire

Now, Young's modulus of a wire of cross-section (A) when some force (F) is applied, $Y = \frac{Fl}{A\Delta l}$

$$k_1 = \frac{Y_1 A_1}{\ell_1} = \frac{120 \times 10^9 A}{\ell}$$

$$k_2 = 60 \times 10^9 \frac{A}{\ell}$$

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

$$F = k_{eq}(x)$$

$$\frac{F}{A} = \frac{k_{eq}}{A}(x) = 8 \times 10^6$$

Solution gives this answer which is not matching with any option.

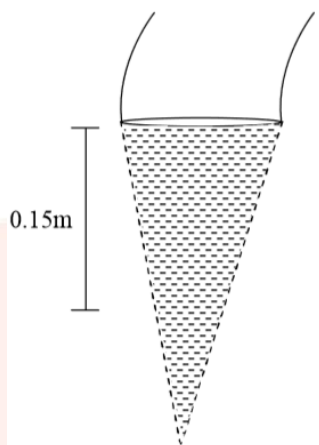
19. Water from a tap emerges vertically downwards with an initial speed of 1.0 ms^{-1} . The cross-sectional area of the tap is 10^{-4} m^2 . Assume that the pressure is constant throughout the stream of water and that the flow is streamlined. The cross-sectional area of the stream, 0.15 m below the tap would be :

(Take $g=10 \text{ ms}^{-2}$)

- A. $1 \times 10^{-5} \text{ m}^2$ B. $2 \times 10^{-5} \text{ m}^2$
C. $5 \times 10^{-4} \text{ m}^2$ D. $5 \times 10^{-5} \text{ m}^2$

Ans. D

Sol:



From equation of continuity.

$$A \propto \frac{1}{V}$$

Where, A = area of flow
and V = velocity of flow.

Now we have

$$A_1 V_1 = A_2 V_2$$

$$10^{-4} \times 1 = A_2 V_2$$

$$A_2 V_2 = 10^{-4}$$

Bernoulli's theorem

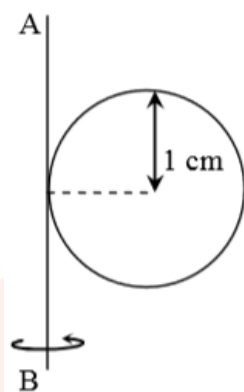
$$\frac{1}{2} \rho (V_1^2 - V_2^2) + \rho gh = 0$$

$$V_2^2 = V_1^2 + 2gh$$

$$V_2 = 2$$

$$\therefore A_2 = \frac{10^{-4}}{2} = 5 \times 10^{-5} \text{ m}^2$$

20. A metal coin of mass 5 g and radius 1 cm is fixed to a thin stick AB of negligible mass as shown in the figure. The system is initially at rest. The constant torque, that will make the system rotate about AB at 25 rotations per second in 5s, is close to:



- A. $4.0 \times 10^{-6} \text{ Nm}$ B. $1.6 \times 10^{-5} \text{ Nm}$
C. $2.0 \times 10^{-5} \text{ Nm}$ D. $7.9 \times 10^{-6} \text{ Nm}$

Ans. C

Sol:

As we know that torque, $\tau = I\alpha$ where

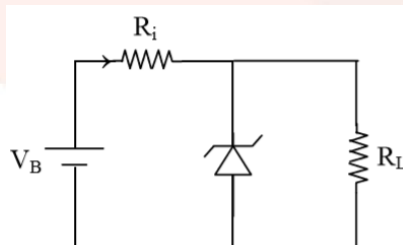
$$\alpha = \frac{d\omega}{dt} = 10\pi \text{ and } I = \frac{5}{4}(MR^2)$$

Torque, $\tau = I\alpha$

$$= \frac{5}{4}(MR^2)\alpha$$

$$= 2 \times 10^{-5} \text{ N-m}$$

21. The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is, $R_L = 4\text{k}\Omega$. The series resistance of the circuit is $R_i = 1\text{k}\Omega$. If the battery voltage V_B varies from 8 V to 16 V, what are the minimum and maximum values of the current through Zener diode?



- A. 0.5 mA; 8.5 mA B. 1.5 mA; 8.5 mA
C. 0.5 mA; 6 mA D. 1 mA; 8.5 mA

Ans. A

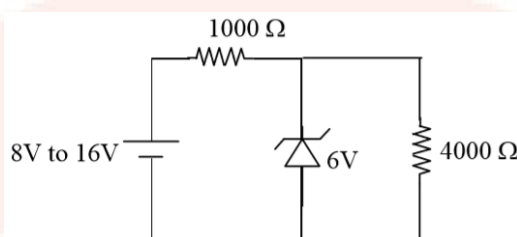
Sol:

Given that

Zener breakdown voltage, $V_z = 6V$

So, current through load resistance is

$$i_L = \frac{V_z}{R_L} = \frac{6}{4 \times 10^3} = 1.5 \text{ mA}$$



Now $V_B = 8 \text{ V}$

total current = 2mA

$$I_z = 2 - 1.5 = 0.5 \text{ mA}$$

Similarly when $V_B = 16V$

$$V_{R_1} = 16 - 6 = 10V$$

$$\therefore i_1 = \frac{10}{1 \times 10^3} = 10 \times 10^{-3} \text{ A} = 10 \text{ mA}$$

total current = 10 mA

$$I_z = 10 - 1.5 = 8.5 \text{ mA}$$

22. Two radioactive substances A and B have decay constants 5λ and λ respectively. At $t = 0$, a sample has the same number of the two nuclei. The time taken for the ratio of the number of nuclei to become $\left(\frac{1}{e}\right)^2$ will be :

A. $2/\lambda$

B. $1/2\lambda$

C. $1/4 \lambda$

D. $1/\lambda$

Ans. B

Sol: Given that,

Decay constant of radioactive substances A = 5λ

Decay constant of radioactive substances B = λ

$$\frac{N_1}{N_2} = \frac{e^{-5\lambda t}}{e^{-\lambda t}} = e^{-4\lambda t} = e^{-2}$$

$$\therefore t = \frac{1}{2\lambda}$$

23. A bullet of mass 20 g has an initial speed of 1 ms^{-1} , just before it starts penetrating a mud wall of thickness 20 cm. IF the wall offers a mean resistance of $2.5 \times 10^{-2} \text{ N}$, the speed of the bullet after emerging from the other side of the wall is close to :

- A. 0.7 ms^{-1} B. 0.3 ms^{-1}
C. 0.1 ms^{-1} D. 0.4 ms^{-1}

Ans. A

Sol: Given that

$$M = 20 \text{ g} = 20 \times 10^{-3} \text{ kg}$$

$$s = 20 \times 10^{-2} \text{ m}$$

$$a = \frac{F}{m} = -\frac{2.5 \times 10^{-2}}{20 \times 10^{-3}} = -1.25 \text{ m/s}^2$$

As we know that

$$v^2 = u^2 + 2as$$

$$(\because u = 1 \text{ ms}^{-1} \text{ and } s = 20 \text{ cm} = 0.2 \text{ m})$$

$$v^2 = 1 - 2 \times 1.25 \times 0.2$$

$$v^2 = 0.5 = \frac{1}{2}$$

$$v = \frac{1}{\sqrt{2}} = 0.7 \text{ m/s}$$

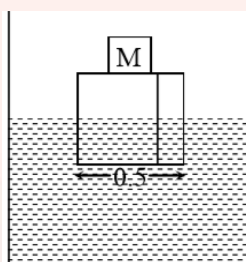
24. A cubical block of side 0.5 m floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water ?

[Take, density of water = 10^3 kg/m^3]

- A. 30.1 kg B. 65.4 kg
C. 87.5 kg D. 46.3 kg

Ans. C

Sol:



By Archimedes' principle,
weight of block = weight of displaced water

$$\Rightarrow V\rho_b g = (30\%V) \rho_w g$$

where ρ_b = density of block,

ρ_w = density of water

and V = volume of block

$$\text{So, } \rho_b = 0.3 \rho_w$$

$$M = \rho_w [0.5 \times 0.5 \times 0.35]$$

$$M = 87.5 \text{ kg}$$

25. A submarine experiences a pressure of $5.05 \times 10^6 \text{ Pa}$ at a depth of d_1 in a sea. When it goes further to a depth of d_2 , it experiences a pressure of $8.08 \times 10^6 \text{ Pa}$. Then $d_2 - d_1$ is approximately (density of water = 10^3 kg/m^3 and acceleration due to gravity = 10 ms^{-2}):

- A. 400 m B. 500 m
C. 300 m D. 600 m

Ans. C

Sol:

Pressure inside a fluid volume open to atmosphere is

$$p = p_0 + h\rho g$$

where, p = pressure at depth h , h = depth,

ρ = density of fluid

and g = acceleration due to gravity.

When $h = d_1$, pressure $p_1 = 5.05 \times 10^6$ Pa and

When $h = d_2$, pressure $p_2 = 8.08 \times 10^6$ Pa

So, we have

$$P_1 = P_0 + \rho g d_1$$

$$P_2 = P_0 + \rho g d_2$$

$$\therefore \Delta P = \rho g [d_2 - d_1] = 3.03 \times 10^6$$

$$d_2 - d_1 = 303 \text{ m}$$

$$= 300 \text{ m}$$

26. Light is incident normally on a completely absorbing surface with an energy flux of 25 Wcm^{-2} . If the surface has an area of 25 cm^2 , the momentum transferred to the surface in 40 min time duration will be:

A. $5.0 \times 10^{-3} \text{ Ns}$ B. $3.5 \times 10^{-6} \text{ Ns}$

C. $6.3 \times 10^{-4} \text{ Ns}$ D. $1.4 \times 10^{-6} \text{ Ns}$

Ans. A

Sol:

Radiation pressure over an absorbing surface is, $p = I/c$

where, I = intensity of energy flux

and c = speed of light.

If A = area of surface, then force due to radiation on the surface is

$$\begin{aligned}\therefore \frac{F}{A} &= \frac{I}{C} \\ F &= \frac{IA}{C} = \frac{\Delta P}{\Delta t} \\ \Delta P &= \frac{I}{C} A \Delta t \\ &= \frac{25 \times 10^4}{3 \times 10^8} \times 25 \times 10^{-4} \times 40 \times 60 \\ &= 5 \times 10^{-3} \text{ N s}\end{aligned}$$

27. In a Young's double slit experiment, the ratio of the slit's width is 4: 1. The ratio of the intensity of maxima to minima, close to the central fringe on the screen, will be:

- A. 25: 9 B. $(\sqrt{3} + 1)^4 : 16$
C. 9: 1 D. 4 : 1

Ans. C

Sol: Given that

$$\begin{aligned}I_1 &= 4I_0 & I_2 &= I_0 \\ I_{\max} &= (\sqrt{I_1} + \sqrt{I_2})^2 = 9I_0 \\ I_{\min} &= (\sqrt{I_1} - \sqrt{I_2})^2 = I_0\end{aligned}$$

In YDSE, ratio of intensities of maxima and minima is given by

$$\begin{aligned}\frac{I_{\max}}{I_{\min}} &= \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} \\ \therefore \frac{I_{\max}}{I_{\min}} &= \frac{9}{1}\end{aligned}$$

28. The time dependence of the position of a particle of mass $m = 2$ is given by $\vec{r}(t) = 2t\hat{i} - 3t^2\hat{j}$. Its angular momentum, with respect to the origin, at time $t = 1$ is :

- A. $-34(\hat{k} - \hat{i})$ B. $48(\hat{i} + \hat{j})$
C. $36\hat{k}$ D. $-48\hat{k}$

Ans. D

Sol:

Position of particle is, $\vec{r} = 2t\hat{i} - 3t^2\hat{j}$

where, t is instantaneous time,

velocity of particle is $\vec{v} = \frac{d\vec{r}}{dt} = 2\hat{i} - 6t\hat{j}$

Now, angular momentum of particle with respect to origin is given by

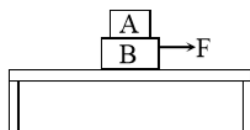
$$\begin{aligned} L &= m(\vec{r} \times \vec{v}) \\ &= m\{2t\hat{i} - 3t^2\hat{j} \times (2\hat{i} - 6t\hat{j})\} \\ \Rightarrow L &= -m(6t^2)\hat{k} \end{aligned}$$

So, angular momentum of particle of mass 2 kg at time $t = 2$ is

$$L = (-2 \times 6 \times 2^2)\hat{k} = -48\hat{k}$$

29. Two blocks A and B of masses $m_A = 1$ kg and $m_B = 3$ kg are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is :

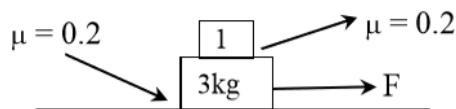
[Take $g = 10 \text{ m/s}^2$]



- A. 16 N B. 40 N
C. 8 N D. 12 N

Ans. A

Sol:



We consider limiting case,

$$m_A a = f_2 \Rightarrow m_A a = \mu(m_A)g$$

$$\Rightarrow a = \mu g = 0.2 \times 10 = 2 \text{ ms}^{-2} \dots(i)$$

Now,

$$a = \frac{F - \mu(m_A + m_B)g}{(m_A + m_B)} = \frac{F - 0.2(1+3) \times 10}{1+3}$$

$$a = \frac{F - 8}{4} \dots(ii)$$

from (i) and (ii)

$$F = 16\text{N}$$

30. A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is :

[Given Planck's constant $h = 6.6 \times 10^{-34}$ Js, speed of light $c = 3.0 \times 10^8$ m/s]

- A. 5×10^{15} B. 1×10^{16}
C. 1.5×10^{16} D. 2×10^{16}

Ans. A

Sol:

Power of laser is given by

$$P = \frac{\text{Energy}}{\text{Time}} = \frac{\text{Number of photons emitted} \times \text{Energy of one photon}}{\text{Time}}$$

$$P = \frac{nhc}{\lambda}$$

$$n = \frac{P\lambda}{hc} = \frac{2 \times 10^{-3} \times 5 \times 10^{-7}}{6.6 \times 10^{-34} \times 3 \times 10^8}$$

$$\approx 5 \times 10^{15}$$

Chemistry

31. 1 g of a non – volatile non – electrolyte solute is dissolved in 100 g of two different solvents A and B whose ebullioscopic constants are in the ratio of 1: 5. The ratio of the elevation in their boiling points, $\frac{\Delta T_b(A)}{\Delta T_b(B)}$, is :

- A. 5: 1 B. 1: 5
C. 10: 1 D. 1: 0.2

Ans. B

Sol:

The expression of elevation of boiling point

$$\Delta T_b = K_b \times m \times i$$

where K_b = ebullioscopic constant

m = molality

i = van't Hoff factor

So, the expression becomes

$$\Delta T_b \propto K_b$$

$$\frac{\Delta T_{b(A)}}{\Delta T_{b(B)}} = \frac{(K_b \times m)_A}{(K_b \times m)_B}$$

Molality is same for both the solution.

$$= \frac{1}{5} \times 1$$

$$= 1 : 5$$

32. The crystal field stabilization energy (CFSE) of $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $\text{K}_2[\text{NiCl}_4]$, respectively, are:

- A. $-0.4\Delta_o$ and $-0.8\Delta_t$ B. $-0.4\Delta_o$ and $-1.2\Delta_t$
C. $-0.6\Delta_o$ and $-0.8\Delta_t$ D. $-2.4\Delta_o$ and $-1.2\Delta_t$

Ans. A

Sol:

As we know that

Crystal field stabilisation energy (CFSE)

For octahedral complexes = $(-0.4x + 0.6y) \Delta_o$

Where x = number of electron occupying t_{2g} orbital

y = number of electrons occupying e_g orbital.

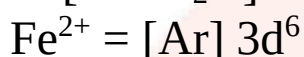
CFSE for tetrahedral complex

$$= (-0.6x + 0.4y) \Delta_t$$

Where x = number of electrons occupying e orbital

y = number of electrons occupying t orbital.

In $[\text{FeCH}_2\text{O}]^{2+}$

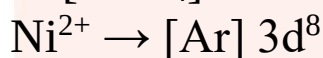


H_2O is a weak field ligand so pairing do not take place

T_{2g}	e_g
2, 1, 1	11

$$\therefore \text{C.F.S.F} = -0.4 \times 4\Delta_o + 0.6 \times 2\Delta_o = -0.4\Delta_o$$

In $[\text{NiCl}_4]^{2-}$



$\text{Cl}^- \rightarrow$ weak field ligand, Δ_o pairing do not takes place and have tetrahedral geometry

e_g	t_{2g}
2, 2	2, 11

$$\text{C.F.S.E} = -0.6 \times 4\Delta_t + 0.4 \times \Delta_t$$

$$= -2.4 \Delta_t + 1.6 \Delta_t$$

$$= -0.8 \Delta_t$$

33. The correct match between Item – I and Item – II is:

	Item – I		Item – II
(a)	High density polythene	(I)	Peroxide catalyst
(b)	Polyacrylonitrile	(II)	Condensation at high temperature & pressure
(c)	Novolac	(III)	Ziegler – Natta Catalyst
(d)	Nylon 6	(IV)	Acid or base catalyst

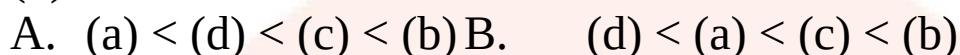
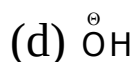
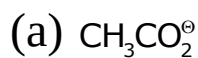
- A. (a) → (III), (b) → (I), (c) → (IV), (d) → (II)
 B. (a) → (IV), (b) → (II), (c) → (I), (d) → (III)
 C. (a) → (III), (b) → (I), (c) → (II), (d) → (IV)
 D. (a) → (II), (b) → (IV), (c) → (I), (d) → (III)

Ans. B

Sol:

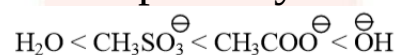
	Item – I		Item – II
(a)	High density polythene	(IV)	Acid or base catalyst
(b)	Polyacrylonitrile	(II)	Condensation at high temperature & pressure
(c)	Novolac	(I)	Peroxide catalyst
(d)	Nylon 6	(III)	Ziegler – Natta Catalyst

34. The increasing order of nucleophilicity of the following nucleophiles is:



Ans. C

Sol: Greater the negative charge Present on a nucleophilic centre greater would be its nucleophilicity. So, nucleophilicity order



35. The correct option among the following is:

A. Colloidal particles in lyophobic sols Can be precipitated by electrophoresis.

B. Colloidal medicines are more effective because they have small surface area.

C. Brownian motion in colloidal solution is faster if the viscosity of the solution is very high.

D. Addition of alum to water makes it unfit for drinking.

Ans. A

Sol: Electrophoresis is used to coagulate lyophobic colloids.

36. The noble gas that does NOT occur in the atmosphere is :

- A. He B. Kr
C. Ra D. Ne

Ans. C

Sol: Radon is not naturally occurring in the atmosphere. Because radon is produced by the radioactive decay of radium – 226

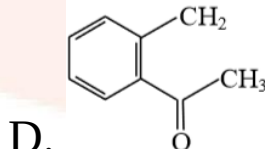
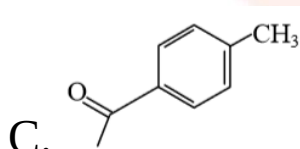
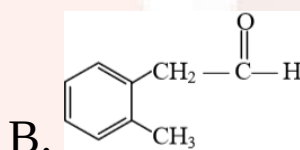
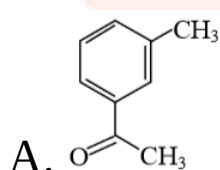
37. The highest possible oxidation states of uranium and plutonium, respectively, are:

- A. 6 and 7 B. 4 and 6
C. 6 and 4 D. 7 and 6

Ans. A

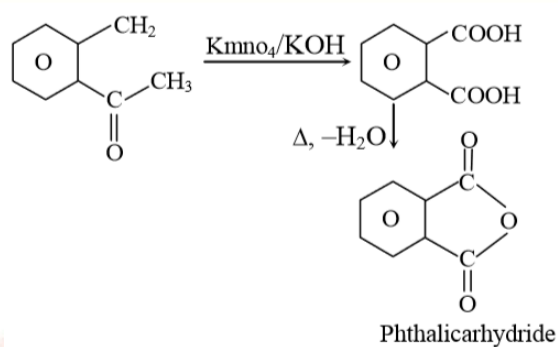
Sol: This highest oxidation state of uranium is +6 whereas of plutonium is + 7

38. Compound A ($C_9H_{10}O$) shows positive iodoform test. Oxidation of A with $KMnO_4/KOH$ gives acid B($C_8H_6O_4$). Anhydride of B is used for the preparation of phenolphthalein. Compound A is:

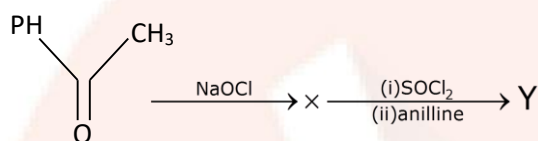


Ans. D

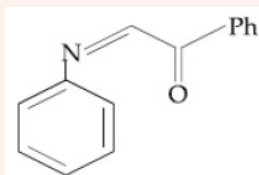
Sol:



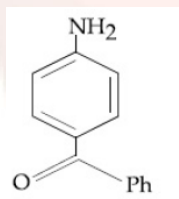
39. The major product 'Y' in the following reaction is:



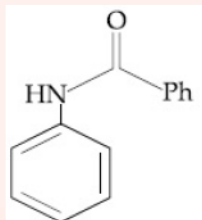
A.



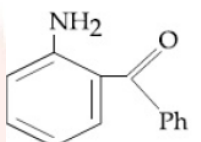
B.



C.

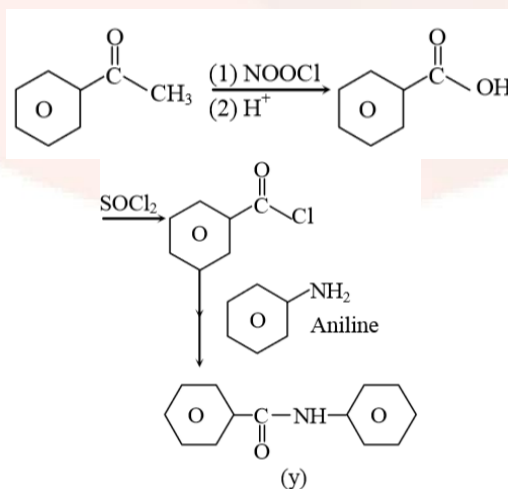


D.



Ans. C

Sol:



40. The number of pentagons in C_{60} and trigons (triangles) in white phosphorus, respectively, are:

- A. 20 and 4 B. 12 and 4
C. 12 and 3 D. 20 and 3

Ans. B

Sol:

In C_{60} (Buckminster fullerene) twenty hexagons and twelve pentagons are presents which are interlocked resulting a shape of soccer ball.

Phosphorous has large atomic size and less electronegativity, so it forms single bond instead of $p\pi-p\pi$ multiple bond.

$\therefore$ No of triangle (triangles) = 4

41. For the reaction of H_2 with I_2 , the rate constant is $2.5 \times 10^{-4} \text{ dm}^3 \text{ mol}^{-1} \text{ S}^{-1}$ at 327° C and $1.0 \text{ dm}^3 \text{ mol}^{-1} \text{ S}^{-1}$ at 527° C . The activation energy for the reaction, in kJ mol^{-1} is:

($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- A. 166 B. 150
C. 59 D. 72

Ans. A

Sol:

The Arrhenius equation for rate constants at two different temperature is

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303k} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

where, k_1 and k_2 are rate constants at temperature T_1 and T_2 , respectively.

R = Gas constant, E_a = Activation energy

$$\log \frac{1}{2.5 \times 10^{-4}} = \frac{E_a}{2.3 \times 8.314} \left(\frac{1}{600} - \frac{1}{800} \right)$$

$$\log_{10} 4 \times 10^3 = \frac{E_a}{2.3 \times 8.3} \left(\frac{1}{2400} \right)$$

$$= \frac{3.6 \times 2.3 \times 8.3 \times 2400}{1000} \text{ kg/mol} = E_a = 164.93$$

42. The correct statement is:

- A. Sodium cyanide cannot be used in the metallurgy of silver.
- B. Zincite is a carbonate ore.
- C. Zone refining process is used for the refining of titanium.
- D. Aniline is a froth stabilizer.

Ans. D.

Sol: Ti is refined by Van Arkel method. Ag is leached by dilute solution of NaCN. Zincite is ZnO. Aniline is a froth stabilizer.

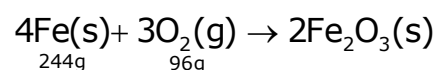
43. The minimum amount of O₂ (g) consumed per gram of reactant is for the reaction:

[Given atomic mass : Fe = 56, O = 16, Mg = 24, P = 31, C = 12, H = 1)

- A. $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
- B. $4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3(\text{s})$
- C. $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$
- D. $\text{P}_4(\text{s}) + 5\text{O}_2(\text{g}) \rightarrow \text{P}_4\text{O}_{10}(\text{s})$

Ans. B

Sol:

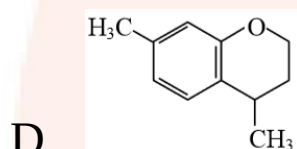
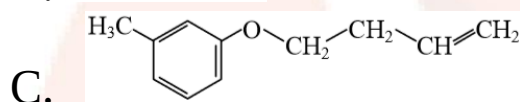
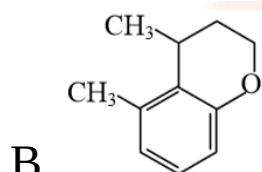
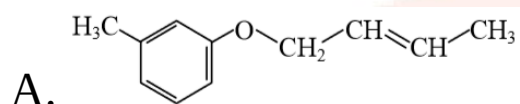
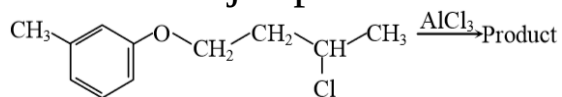


$\Rightarrow 1\text{g of reactant} \frac{96}{224} \text{ g of O}_2 \text{ consumed}$

0.43g

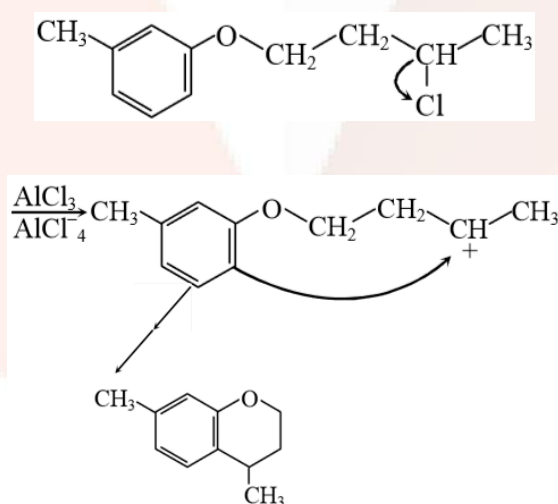
So, minimum amount O_2 is consumed per gram of reactant (Fe) in reaction (b).

44. The major product obtained in the given reaction is :



Ans. D

Sol: It is eq. of friedel craft alkylation reaction.



45. The correct statements among (a) to (d) are:

- (a) saline hydrides produce H_2 gas when reacted with H_2O .
 - (b) reaction of $LiAlH_4$ with BF_3 leads to B_2H_6
 - (c) PH_3 and CH_4 are electron – rich and electron – precise hydrides, respectively.
 - (d) HF and CH_4 are called as molecular hydrides.
- A. (c) and (d) only B. (a), (b), (c) and (d).
C. (a), (c) and (d) only. D. (a), (b) and (c) only

Ans. B

Sol: Statement a, b, c and d are the correct statements as per the facts.

46. The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are:

- A. Balmer and Brackett
- B. Paschen and Pfund
- C. Brackett and Pfund
- D. Lyman and Paschen

Ans. D

Sol:

According to Rydberg's equation

$$\frac{1}{\lambda} = \frac{R_H}{hc} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ and } \frac{12400}{\Delta E(\text{eV})}$$

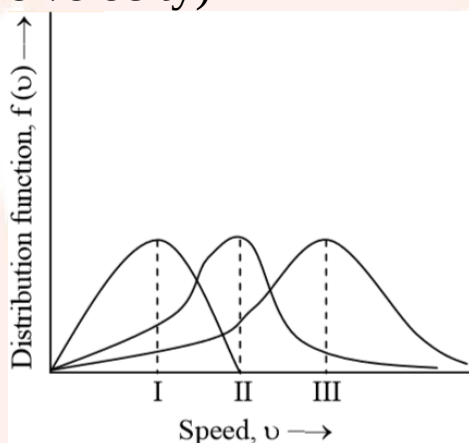
Note: Lyman = L ($n_1 = 1$) Balmer = B ($n_1 = 2$) Paschen = P ($n_1 = 3$), Brackett = Bk ($n_1 = 4$) Pfund = Pf ($n_1 = 5$).

Ratio of shortest wavelength of two spectral series wavelength shortest means energy maximum.

$$\frac{(\Delta E_{\text{lyman}})_{\text{max}}}{(\Delta E_{\text{pascmen}})_{\text{max}}} = \frac{13.6 \times 2^2 \left(\frac{1}{2} - \frac{1}{\infty} \right)}{13.6 \times 2^2 \left(\frac{1}{9} - \frac{1}{\infty} \right)}$$

$$\frac{(\Delta E_{\text{lyman}})_{\text{max}}}{(\Delta E_{\text{pascmen}})_{\text{max}}} = \frac{9}{1}$$

47. Points I, II and III in the following plot respectively correspond to
(V_{mp} : most probable velocity)



- A. V_{mp} of N_2 (300 K); V_{mp} of O_2 (400K); V_{mp} of H_2 (300 K)
- B. V_{mp} of O_2 (400 K); V_{mp} of N_2 (300K); V_{mp} of H_2 (300 K)
- C. V_{mp} of N_2 (300 K); V_{mp} of H_2 (300K); V_{mp} of O_2 (400 K)
- D. V_{mp} of H_2 (300 K); V_{mp} of N_2 (300K); V_{mp} of O_2 (400 K)

Ans. A

Sol:

From kinetic gas equation,

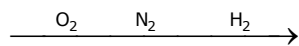
$$\text{Most probable velocity } (v_{\text{mp}}) = \sqrt{\frac{2RT}{M}}$$

where, R = gas constant, T = temperature

M = molecular mass

$$V_{mp} = \sqrt{\frac{2RT}{m}}$$

$$V_{mp} \propto \frac{1}{\sqrt{m}}$$



m decreases

So, V_{mp} $\longrightarrow$ increases

$$V_{mp} \text{ of O}_2 = \sqrt{\frac{2 \times R \times 400}{32}}$$

$$V_{mp} \text{ of N}_2 = \sqrt{\frac{2 \times R \times 300}{28}}$$

$$V_{mp} \text{ of H}_2 = \sqrt{\frac{2 \times R \times 300}{2}}$$

$$V_{mp \text{ I}} \text{ of O}_2 < V_{mp \text{ II}} \text{ of N}_2 < V_{mp \text{ III}} \text{ of H}_2$$

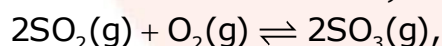
48. The correct order of the first ionization enthalpies is:

- A. $\text{Mn} < \text{Ti} < \text{Zn} < \text{Ni}$ B. $\text{Zn} < \text{Ni} < \text{Mn} < \text{Ti}$
 C. $\text{Ti} < \text{Mn} < \text{Ni} < \text{Zn}$ D. $\text{Ti} < \text{Mn} < \text{Zn} < \text{Ni}$

Ans. C

Sol: The correct order of the first ionization enthalpies is
 $\text{Zn} > \text{Ni} > \text{Mn} > \text{Ti}$

49. For the reaction,



$$\Delta H = -57.2 \text{ kJ mol}^{-1} \text{ and}$$

$$K_c = 1.7 \times 10^{16}$$

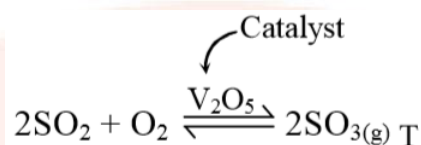
Which of the following statement is INCORRECT?

- A. The equilibrium will shift in forward direction as the pressure increases.
 B. The equilibrium constant is large suggestive of reaction going to completion and so no catalyst is required.

- C. The equilibrium constant decreases as the temperature increases.
- D. The addition of inert gas at constant volume will not affect the equilibrium constant

Ans. B

Sol:

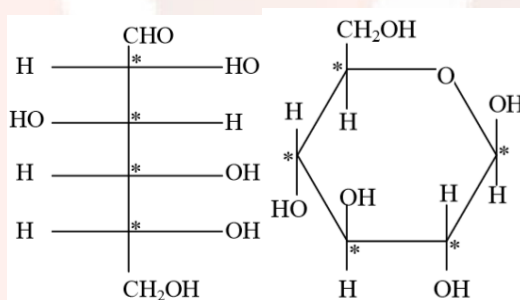


50. Number of stereo centers present in linear and cyclic structures of glucose are respectively:

- A. 5 & 4 B. 5 & 5
C. 4 & 4 D. 4 & 5

Ans. D

Sol:

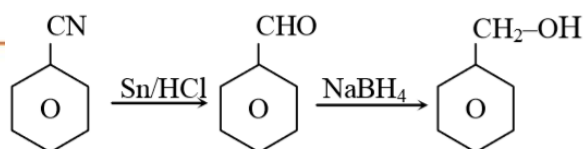


51. Which of the following is NOT a correct method of the preparation of benzy lamine from cyanobenzene ?

- A. H_2 / Ni
B. (i) $\text{SnCl}_2 + \text{HCl}(\text{gas})$ (ii) NaBH_4
C. (i) $\text{HCl}/\text{H}_2\text{O}$ (ii) NaBH_4
D. (i) LiAlH_4 (ii) H_3O^+

Ans. C

Sol:



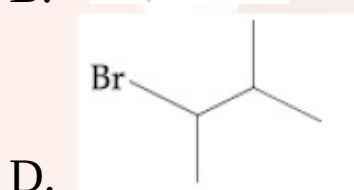
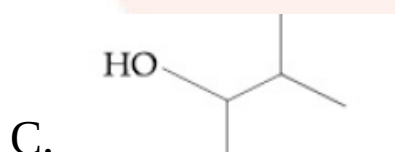
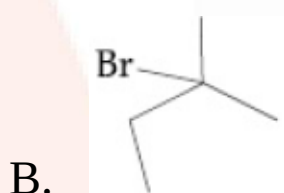
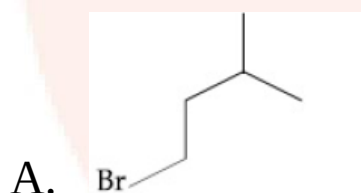
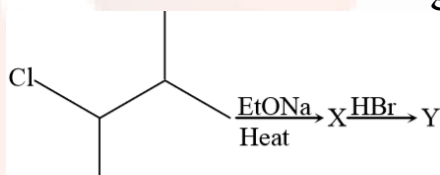
52. In chromatography, which of the following statements is INCORRECT for R_f ?

- A. The value of R_f can not be more than one.
- B. Higher R_f value means higher adsorption.
- C. Higher R_f value depends on the type of
- D. R_f value is dependent on the mobile phase.

Ans. B

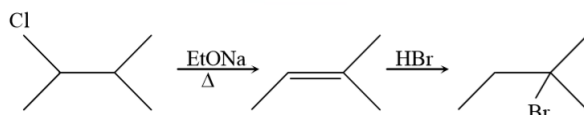
Sol: Low polarity compounds are weakly absorbed and has greater R_f value.

53. The major product 'Y' in the following reaction is :



Ans. B

Sol:



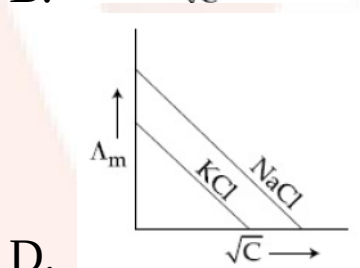
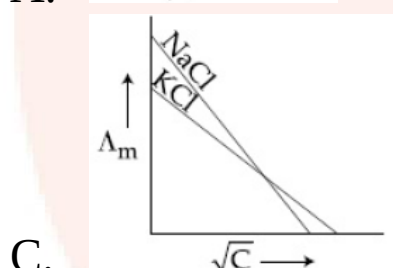
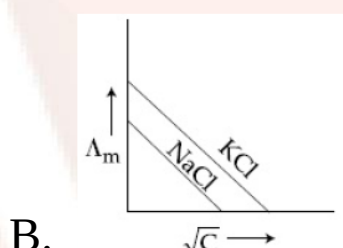
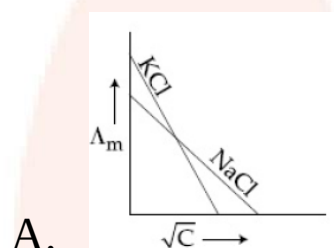
54. Air pollution that occurs in sunlight is :

- A. Frog B. Reducing smog
C. Acid rain D. Oxidising smog

Ans. D

Sol: A/C to the fact that Oxidizing smog form when the primary pollutants are transformed through photochemical reaction into secondary pollutants the most important of which are oxidant gases, ozone and peroxyacetyl nitrate.

55. Which of the following graphs between molar conductivity (Λ_m) versus $\sqrt{C}$ is correct?



Ans. B

Sol:

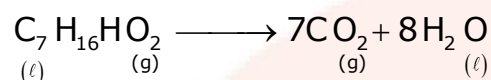
At infinite dilution molar conductivity of KCl is greater than the molar conductivity of NaCl.

56. The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptanes (1) is carried out at a temperature T , is equal to :

- A. $-3 RT$ B. $4 RT$
C. $3 RT$ D. $-4 RT$

Ans. D

Sol:



$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta H - \Delta U = (7 - 11) RT = - 4RT$$

57. Which of these factors does not govern the stability of a conformation in acyclic compounds?

- A. Torsional strain B. Angle strain
C. Steric interactions D. Electrostatic forces of interaction

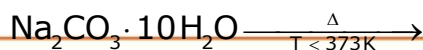
Ans. B

Angle strain is not present in acyclic compounds.

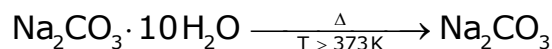
58. A hydrated solid X on heating initially gives a monohydrated compound Y. Y upon heating above 373 K leads to an anhydrous white powder Z. X and Z, respectively, are :

- A. Washing soda and soda ash
B. Baking soda and dead burnt plaster.
C. Washing soda and dead burnt plaster.
D. Baking soda and soda ash

Ans. A



(X)



(Y)

(Z)

X = washing soda

Z = soda ash

59. The INCORRECT statement is:-

- A. The spin –only magnetic moment of $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^2$ is 283 BM.
- B. The spin –only magnetic moments of $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ are nearly similar.
- C. The color of $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ is violet as it absorbs the yellow light.
- D. The gemstone, ruby, has Cr^{3+} ions occupying the octahedral sites of beryl.

Ans. D

Sol:

(D) Ruby, a pink or blood-red coloured gemstone belongs to corundum (Al_2O_3 , alumina) system which has trigonal crystalline lattice containing the repeating unit of $\text{Al}_2\text{O}_3 - \text{Cr}^{3+}$. So, ruby does not belong to beryl lattice ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$).

Thus, statement (D) is Correct.

60. The pH of a 0.02M (NH₄OH)=10⁻⁵ and log 2 = 0.301]

- A. 2.65 B. 4.65
C. 4.35 D. 5.35

Ans. D

Sol:

pH of 0.02 mNH₄Cl

$$\begin{aligned} \text{pH} &= 7 - \frac{P}{2} - \frac{\log C}{2} \\ &= 7 - \frac{5}{2} - \frac{\log^2 \times 10^{-2}}{2} \\ &= 7 - 2.5 + \frac{1.7}{2} \\ &= 4.50 + 0.85 \\ &= 5.35 \end{aligned}$$

Mathematics

61. The smallest natural number n, such that the coefficient of x in the expansion of

$\left(x^2 + \frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is :

- A. 38 B. 23
C. 58 D. 35

Ans. A

Sol:

Given binomial is

$\left(x^2 + \frac{1}{x^3}\right)^n$, its $(r + 1)^{\text{th}}$ term, is

General term

$$T_{r+1} = {}^nC_r x^{2n-2r} \cdot x^{-3r}$$

$$\therefore 2n - 5r = 1 \Rightarrow 2n = 5r + 1$$

$$\therefore r = \frac{2n-1}{5}$$

$$\therefore \text{Coeff. of } x = {}^nC_{\left(\frac{2n-1}{5}\right)} = {}^nC_{23}$$

$$\therefore \frac{2n-1}{5} = 23 \text{ or } n - \left(\frac{2n-1}{5}\right) = 23$$

$$2n - 1 = 115$$

$$n = 38$$

62. If $5x + 9 = 0$ is the directrix of the hyperbola $16x^2 - 9y^2 = 144$, then its corresponding focus is:

A. $\left(\frac{5}{3}, 0\right)$

B. $(-5, 0)$

C. $\left(-\frac{5}{3}, 0\right)$

D. $(5, 0)$

Ans. B

Sol:

Equation of given hyperbola is

$$16x^2 - 9y^2 = 144$$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$a = 3$$

$$b = 4$$

So, the eccentricity of above equation

$$e^2 = 1 + \frac{16}{9}$$

$$e = \frac{5}{3}$$

$$\therefore \text{ focus is } (-ae, 0) = (-5, 0)$$

63. The angles A, B and C of a triangle ABC are in A.P. and $a : b = 1 : \sqrt{3}$. If $c = 4$ cm, then the area (in sq.cm) of this triangle is :

A. $4\sqrt{3}$

B. $\frac{4}{\sqrt{3}}$

C. $\frac{2}{\sqrt{3}}$

D. $2\sqrt{3}$

Ans. D

Sol:

It is given that the angles of a ΔABC are in AP

$$\therefore 2B = A + C$$

As we know sum of all angles of a triangle = π

$$\& A + B + C = \pi$$

$$B = \frac{\pi}{3}$$

$$\therefore A + C = \frac{2\pi}{3}$$

$$\frac{a}{b} = \frac{1}{\sqrt{3}} \quad \text{Given}$$

$$\frac{2R \sin A}{2R \sin B} = \frac{1}{\sqrt{3}}$$

$$\sin A = \frac{1}{2}$$

$$\therefore A = 30^\circ$$

$$\therefore a = 2, b = 2\sqrt{3}, c = 4$$

$$\text{Area of } \Delta ABC = \frac{1}{2} ab \sin C$$

$$\Delta = \frac{1}{2} \times 2\sqrt{3} \times 2 \times 1 = 2\sqrt{3}$$

64. If the tangent to the curve $y = \frac{x}{x^2 - 3}$, $x \in \mathbb{R}, (x \neq \pm\sqrt{3})$, at a point $(\alpha, \beta) \neq (0, 0)$ on it is parallel to the line $2x + 6y - 11 = 0$, then :

- A. $|6\alpha + 2\beta| = 19$ B. $|2\alpha + 6\beta| = 19$
C. $|2\alpha + 6\beta| = 11$ D. $|6\alpha + 2\beta| = 9$

Ans. B

Sol:

Equation of given curve is

$$y = \frac{x}{x^2 - 3}, x \in \mathbb{R}, (x \neq \pm\sqrt{3}) \quad \dots(i)$$

On differentiating eq. (i) w.r.t. x, we get

$$\frac{dy}{dx} = \frac{(x^2 - 3) - x(2x)}{(x^2 - 3)^2} = \frac{(-x^2 - 3)}{(x^2 - 3)^2}$$

It is given that tangent at a point $(\alpha, \beta) \neq (0, 0)$ on it is parallel to the line

$$2x + 6y - 11 = 0$$

$$\therefore \text{Slope of this line} = -\frac{2}{6} = \frac{dy}{dx}\bigg|_{(\alpha, \beta)}$$

$$\frac{dy}{dx}\bigg|_{(\alpha, \beta)} = \frac{-\alpha^2 - 3}{(\alpha^2 - 3)^2}$$

Given

$$\frac{-\alpha^2 - 3}{(\alpha^2 - 3)^2} = -\frac{1}{3}$$

$$\Rightarrow \alpha = 0, \pm 3 (\alpha \neq 0)$$

Now, from equation (i)

$$\beta = \frac{\alpha}{\alpha^2 - 3} = \frac{1}{2} \text{ or } -\frac{1}{2}$$

$$\text{then } |6\alpha + 2\beta| = 19$$

65. Minimum number of times a fair coin must be tossed so that the probability of getting at least one head is more than 99% is :

- A. 8 B. 6
C. 5 D. 7

Ans. D

Sol:

Now, let n be the minim numbers of toss required to get at least one head, then required probability = $1 - (\text{probability that on all 'n' toss we are getting tail})$

$$1 - \left(\frac{1}{2}\right)^n$$

According to questions

$$1 - \left(\frac{1}{2}\right)^n > \frac{99}{100}$$

$$\left(\frac{1}{2}\right)^n < \frac{1}{100}$$

$$\Rightarrow n = 7$$

66. The negation of the Boolean expression $\sim s \vee (\sim r \wedge s)$ is equivalent to :

- A. $\sim s \wedge \sim r$ B. r
C. $s \wedge r$ D. $s \vee r$

Ans. C

Sol:

The given Boolean expression is $\sim s \vee ((\sim r) \wedge s)$

Now, the negation of given Boolean expression is

$$\sim (\sim s \vee (\sim r \wedge s))$$

$$s \wedge (r \vee \sim s)$$

$$(s \wedge r) \vee (s \wedge \sim s)$$

$$(s \wedge r) \wedge (s)$$

$$(s \wedge r)$$

67. The number of real roots of the equation $5 + |2^x - 1| = 2^x(2^x - 2)$

is :

A. 3

B. 1

C. 2

D. 4

Ans. B

Sol:

Given that

$$5 + |2^x - 1| = 2^x(2^x - 2)$$

Case I

$$2^x \geq 1$$

$$5 + 2^x - 1 = 2^x(2^x - 2)$$

$$\text{Let } 2^x = t$$

$$5 + t - 1 = t(t - 2)$$

$$t = 4, -1 \text{ (rejected)}$$

$$2^x = 4$$

$$x = 2$$

only 1 solution

Case II

$$2^x < 1$$

$$5 + 1 - 2^x = 2^x(2^x - 2)$$

$$2^x = t$$

$$5 + 1 - t = t(t - 2)$$

$$0 = t^2 - t - 6$$

$$0 = (t - 3)(t - 2)$$

$$t = 3, -2$$

$$2^x = 3, 2^x = -2$$

(rejected)

Therefore, number of real roots is one.

68. The tangent and normal to the ellipse $3x^2 + 5y^2 = 32$ at the point P(2,2) meet the x-axis at Q and R, respectively. Then the area (in sq. units) of the triangle PQR is :

- A. $\frac{68}{15}$ B. $\frac{34}{15}$
C. $\frac{16}{3}$ D. $\frac{14}{3}$

Ans. A

Sol:

The equation of given ellipse is

$$3x^2 + 5y^2 = 32$$

On differentiating both side of above equation we get

$$6x + 10yy' = 0$$

$$y' = \frac{-3x}{5y}$$

$$y'_{(2,2)} = -\frac{3}{5}$$

$$\text{Tangent } (y - 2) = -\frac{3}{5}(x - 2)$$

$$\Rightarrow Q\left(\frac{16}{3}, 0\right)$$

$$\text{Normal } (y - 2) = \frac{5}{3}(x - 2)$$

$$\Rightarrow R\left(\frac{4}{5}, 0\right)$$

$$\text{Area} = \frac{1}{2}(\text{QR}) \times 2 = \text{QR} = \frac{68}{15}$$

$$\left[\because \text{QR} = \sqrt{\left(\frac{16}{3} - \frac{4}{5}\right)^2} = \sqrt{\left(\frac{68}{15}\right)^2} = \frac{68}{15} \text{ and height} = 2 \right]$$

69. Let λ be a real number for which the system of linear equations

$$x + y + z = 6$$

$$4x + \lambda y - \lambda z = \lambda - 2$$

$$3x + 2y - 4z = -5$$

has infinitely many solutions. Then λ is a root of the quadratic equation :

A. $\lambda^2 + \lambda - 6 = 0$

B. $\lambda^2 - \lambda - 6 = 0$

C. $\lambda^2 - 3\lambda - 4 = 0$

D. $\lambda^2 + 3\lambda - 4 = 0$

Ans. B

Sol:

Given, system of linear equations

$$x + y + z = 6 \quad \dots(i)$$

$$4x + \lambda y - \lambda z = \lambda - 2 \quad \dots(ii)$$

$$\text{and } 3x + 2y - 4z = -5 \quad \dots(iii)$$

has infinitely means solutions, then

$$\Delta = 0$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 4 & \lambda & -\lambda \\ 3 & 2 & -4 \end{vmatrix} = 0$$

$$\Rightarrow -8\lambda + 24 = 0 \quad \Rightarrow \lambda = 3$$

On solving we get $\lambda = 3$

$\therefore$ For $\lambda = 3$, infinitely many solutions is obtained.

70. A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane $x + y + z = 3$ such that the foot of the perpendicular Q also lies on the plane $x - y + z = 3$. Then the co-ordinates of Q are :

A. (2,0,1)

B. (4,0,-1)

C. (1,0,2)

D. (-1,0,4)

Ans. A

Sol:

Let a general point on the given line

$$\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1} = \lambda$$

Let a point on the line is

$$(2\lambda + 1, -\lambda - 1, +\lambda)$$

the foot of perpendicular $Q(x_2, y_2, z_2)$ drawn from point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{z_2 - z_1}{c} = -\frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$$

Foot of $\perp^r$ Q is given by

$$\frac{x - 2\lambda - 1}{1} = \frac{y + \lambda + 1}{1} = \frac{z - \lambda}{1} = -\frac{(2\lambda - 3)}{3}$$

$\therefore$ Q lies on $x + y + z = 3$ and $x - y + z = 3$

$$\Rightarrow x + z = 3 \text{ and } y = 0$$

$$\therefore y = 0 \Rightarrow \lambda + 1 = \frac{-2\lambda + 3}{3} \Rightarrow \lambda = 0$$

$$\therefore Q \text{ is } (2, 0, 1)$$

71. Let a , b and c be in G.P. with common ratio r , where $a \neq 0$ and $0 < r \leq \frac{1}{2}$. If $3a, 7b$ and $15c$ are the first three terms of an A.P., then the 4th term of this A.P. is :

A. $\frac{2}{3}a$ B. $\frac{7}{3}a$

C. a D. $5a$

Ans. C

Sol:

It is given that, the terms a , b , c are in GP with common ratio r , where $a \neq 0$ and $0 < r \leq \frac{1}{2}$.

So, let, $b = ar$ and $c = ar^2$

Now, the terms $3a$, $7b$ and $15c$ are the first three terms of an AP, then

$$14b = 3a + 15c$$

$$14(ar) = 3a + 15(ar^2)$$

$$15r^2 - 14r + 3 = 0$$

$$\Rightarrow r = \frac{1}{3}, \frac{3}{5} \text{ (rejected)}$$

$$\text{Common difference} = 7b - 3a$$

$$= 7ar - 3a$$

$$= \frac{7a}{3} - 3a$$

$$= -\frac{2}{3}a$$

4th term is

$$\Rightarrow 15c - \frac{2}{3}a = \frac{15}{9}a - \frac{2}{3}a = a$$

72. The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is :

A. $\log_e 2 + \frac{3}{2}$

B. $\frac{3}{2} - \frac{1}{\log_e 2}$

C. $\frac{1}{2}$

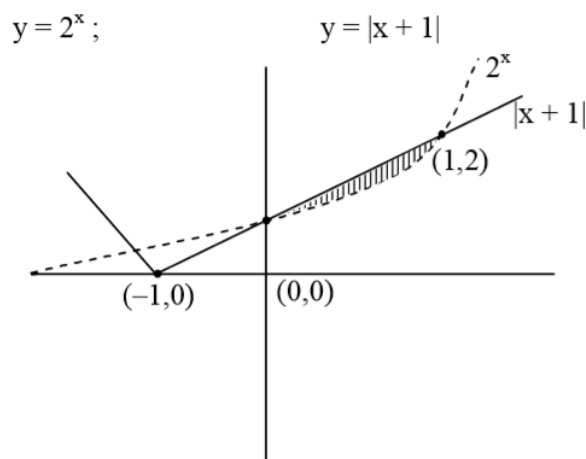
D. $\frac{3}{2}$

Ans. B

Sol:

Given, equation of curves

$$y = 2^x \text{ and } y = |x + 1| = \begin{cases} x+1, & x \geq -1 \\ -x-1, & x < -1 \end{cases}$$



Intersect points (1, 2)

$$\begin{aligned}\text{Required Area} &= \int_0^1 ((x + 1) - 2^x) dx \\ &= \left(\frac{x^2}{2} + x - \frac{2^x}{\log_e 2} \right)_0^1 \\ &= \left[\frac{1}{2} + 1 - \frac{2}{\log_e 2} + \frac{1}{\log_e 2} \right] \\ &= \frac{3}{2} - \frac{1}{\log_e 2}\end{aligned}$$

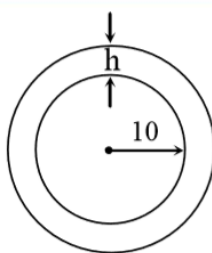
73. A spherical iron ball of radius 10 cm is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of the ice is 5 cm, then the rate at which the thickness (in cm/min) of the ice decreases, is :

- | | |
|---------------------|----------------------|
| A. $\frac{5}{6\pi}$ | B. $\frac{1}{18\pi}$ |
| C. $\frac{1}{9\pi}$ | D. $\frac{1}{36\pi}$ |

Ans. B

Sol:

Let the thickness of layer of ice is x cm, the volume of spherical ball (only ice layer) is



$$V = \frac{4}{3} \pi ((10 + h)^3 - 10^3)$$

On differentiating with respect to x

$$\frac{dV}{dt} = 4\pi(10 + h)^2 \frac{dh}{dt}$$

$$-50 = 4\pi(10 + 5)^2 \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = \frac{-1}{18\pi} \frac{\text{cm}}{\text{min}}$$

74. If $\int x^5 e^{-x^2} dx = g(x)e^{-x^2} + c$, where c is a constant of integration, then $g(-1)$ is equal to :

A. 1

B. $-\frac{5}{2}$

C. $-\frac{1}{2}$

D. -1

Ans. B

Sol:

Given integral, $I = \int x^5 e^{-x^2} dx$

Let $x^2 = t$

$$\Rightarrow \frac{1}{2} \int t^2 e^{-t} dt$$

$$= \frac{1}{2} \left[-t^2 e^{-t} + \int 2te^{-t} dt \right]$$

$$= \frac{-t^2 e^{-t}}{2} - te^{-t} - e^{-t}$$

$$= \left(-\frac{x^4}{2} - x^2 - 1 \right) e^{-x^2} + c \dots (i)$$

∴ It is given that,

$$I = \int x^5 e^{-x^2} dx = g(x) \cdot e^{-x^2} + C$$

By Eq. (i), comparing both sides, we get

$$g(x) = -\frac{1}{2}(x^4 + 2x^2 + 2)$$

$$\text{So, } g(-1) = -\frac{1}{2}(1 + 2 + 2) = -\frac{5}{2}$$

$$g(x) = -\frac{x^4}{2} - x^2 - 1$$

$$g(-1) = -\frac{1}{2} - 1 - 1$$

$$= -\frac{5}{2}$$

75. Suppose that 20 pillars of the same height have been erected along the boundary of a circular stadium. If the top of each pillar has been connected by beams with the top of all its non-adjacent pillars, then the total number of beams is:

- | | |
|--------|--------|
| A. 180 | B. 210 |
| C. 170 | D. 190 |

Ans. C

Sol:

$$\text{Required number of beams} = {}^{20}C_2 - 20 = 190 - 20 = 170$$

76. If $\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$, then $a+b$ is equal to:

- | | |
|-------|-------|
| A. -7 | B. 1 |
| C. 5 | D. -4 |

Ans. A

Sol:

Given that

$$\lim_{x \rightarrow 1} \frac{x^2 - ax + b}{x - 1} = 5$$

Since, limit exist and equal to 5 and denominator is zero at $x = 1$, so numerator $x^2 - ax + b$ should be zero at $x = 1$

$$\text{So, } 1 - a + b = 0 \Rightarrow a = 1 + b \quad \dots (i)$$

Now,

'L' hospital rule

$$2x - a = 5$$

$$2 - a = 5 \quad (\because x = 1)$$

$$a = -3 \quad \dots (ii)$$

Put in (1)

$$\therefore b = -4$$

$$a + b = -7$$

77. Let $a_1, a_2, a_3, \dots$ be an A.P. when $a_6 = 2$. Then the common difference of this A.P., which maximises the product $a_1 a_4 a_5$, is:

A. $\frac{8}{5}$

B. $\frac{2}{3}$

C. $\frac{6}{5}$

D. $\frac{3}{2}$

Ans. A

Sol:

Given, the terms $a_1, a_2, a_3, \dots$ are in AP. Let the common difference of this AP is 'd' and first term $a_1 = a$, then

Common difference = d

$$\therefore a + 5d = 2$$

$$a_1 \cdot a_4 \cdot a_5 = a(a + 3d)(a + 4d)$$

$$f(d) = (2 - 5d)(2 - 2d)(2 - d)$$

$$f'(d) = 0 \Rightarrow d = \frac{2}{3}, \frac{8}{5}$$

$$f''(d) < 0 \text{ at } d = \frac{8}{5}$$

$$\Rightarrow d = \frac{8}{5}$$

78. Let $y = y(x)$ be the solution of the differential equation, $\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that $y(0) = 1$. Then :

A. $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$

B. $y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \sqrt{2}$

C. $y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = -\sqrt{2}$

D. $y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$

Ans. D

Sol:

Given differential equation is

$$\frac{dy}{dx} + y(\tan x) = 2x + x^2 \tan x$$

which is linear differential equation in the form of $\frac{dy}{dx} + Py = Q$.

Here, $P = \tan x$ and $Q = 2x + x^2 \tan x$.

$$\text{I.F.} = e^{\int \tan x \, dx} = e^{\ln \sec x} = \sec x$$

$$y \cdot \sec x = \int (2x + x^2 \tan x) \sec x \, dx$$

$$y \sec x = x^2 \sec x + \lambda$$

$$\Rightarrow y = x^2 + \lambda \cos x$$

$$y(0) = 0 + \lambda = 1$$

$$\Rightarrow \lambda = 1$$

$$y = x^2 + \cos x$$

$$y' = 2x - \sin x$$

$$y'\left(\frac{\pi}{4}\right) = \frac{\pi}{2} - \frac{1}{\sqrt{2}}$$

$$y'\left(-\frac{\pi}{4}\right) = -\frac{\pi}{2} + \frac{1}{\sqrt{2}}$$

79. The integral $\int_{\pi/6}^{\pi/3} \sec^{2/3} x \operatorname{cosec}^{4/3} x dx$ is equal to:

- A. $3^{5/3} - 3^{1/3}$ B. $3^{7/6} - 3^{5/6}$
C. $3^{4/3} - 3^{1/3}$ D. $3^{5/6} - 3^{2/3}$

Ans. B

Sol:

Let

$$I = \int_{\pi/6}^{\pi/3} \sec^{2/3} x \operatorname{cosec}^{4/3} x dx$$

$$\int_{\pi/6}^{\pi/3} \frac{1}{\cos^{2/3} x \sin^{4/3} x} dx$$

$$= \int \frac{\sec^2 x}{\tan^{4/3} x} dx$$

Let $\tan x = t$, $\sec^2 x dx = dt$

$$= \int \frac{dt}{t^{4/3}}$$

$$I = -3(t^{-1/3})$$

$$= \left(-3(\tan x)^{-1/3} \right)_{\pi/6}^{\pi/3}$$

$$= 3 \left(3^{1/3} - \frac{1}{3^{1/6}} \right)$$

$$= 3^{7/6} - 3^{5/6}$$

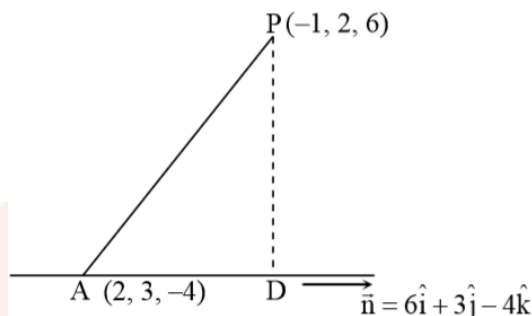
80. The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line passing through the point $(2, 3, -4)$ and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\hat{k}$ is:

- A. 7 B. 6
C. $2\sqrt{13}$ D. $4\sqrt{3}$

Ans. A

Sol:

Let point P whose position vector is $(-\hat{i} + 2\hat{j} + 6\hat{k})$ and a straight line passing through A(2, 3, -4) parallel to the vector $\vec{n} = 6\hat{i} + 3\hat{j} - 4\hat{k}$.



$\therefore$ Required distance = projection of line segment AP perpendicular to vector $\vec{n}$.

$$AD = \left| \frac{\vec{AP} \cdot \vec{n}}{|\vec{n}|} \right| = \sqrt{61}$$

$$\begin{aligned} PD &= \sqrt{AP^2 - AD^2} \\ &= \sqrt{110 - 61} \\ &= 7 \end{aligned}$$

81. Lines are drawn parallel to the line $4x - 3y + 2 = 0$, at a distance $\frac{3}{5}$ from the origin. Then which one of the following points lies on any of these lines?

A. $\left(\frac{1}{4}, \frac{1}{3}\right)$

B. $\left(\frac{1}{4}, -\frac{1}{3}\right)$

C. $\left(-\frac{1}{4}, \frac{2}{3}\right)$

D. $\left(-\frac{1}{4}, -\frac{2}{3}\right)$

Ans. C

Sol:

Since, equation of line parallel to line $ax + by + c = 0$ is $ax + by + \lambda = 0$

$\therefore$ Equation of line parallel to line

$4x - 3y + 2 = 0$ is $4x - 3y + \lambda = 0 \dots(i)$

Now, distance of line (i) from the origin is

$$\frac{|\lambda|}{5} = \frac{3}{5}$$

$$\lambda = \pm 3$$

$\therefore$ required lines are $4x - 3y + 3 = 0$ & $4x - 3y - 3 = 0$

Now, from the given option the point $\left(-\frac{1}{4}, \frac{2}{3}\right)$ lies on the line $4x - 3y + 3 = 0$.

82. Let $f(x) = \log_e(\sin x)$, $(0 < x < \pi)$ and $g(x) = \sin^{-1}(e^{-x})$, $(x \geq 0)$. If α is a positive real number such that $a = (f \circ g)'(\alpha)$ and $b = (f \circ g)(\alpha)$, then:

A. $a\alpha^2 + b\alpha + a = 0$

B. $a\alpha^2 + b\alpha - a = -2\alpha^2$

C. $a\alpha^2 - b\alpha - a = 1$

D. $a\alpha^2 - b\alpha - a = 0$

Ans. C

Sol: Given that

$$f(x) = \log_e(\sin x), (0 < x < \pi) \text{ and } g(x) = \sin^{-1}(e^{-x}), (x \geq 0)$$

$$f(g(x)) = -x \Rightarrow (f(g(x)))' = -1$$

$$f(g(\alpha)) = -\alpha = b \Rightarrow (f(g(\alpha)))' = -1 = a$$

$$\therefore b = -\alpha$$

$$a = -1$$

Since, the value of $a = -1$ and $b = -a$, satisfy the quadratic equation (from the given options)

$$a\alpha^2 - b\alpha = a = 1.$$

83. If the line $ax + y = c$, touches both the curves $x^2 + y^2 = 1$ and $y^2 = 4\sqrt{2}x$, then $|c|$ is equal to :

- A. $\frac{1}{2}$ B. $\frac{1}{\sqrt{2}}$
C. $\sqrt{2}$ D. 2

Ans. C

Sol:

The equation of tangent of slope 'm' to the parabola $y^2 = 4ax$ is $y = mx + \frac{a}{m}$ and a line $ax + by + c = 0$ touches the circle

$$x^2 + y^2 = r^2, \text{ if } \frac{|c|}{\sqrt{a^2 + b^2}} = r.$$

Tangent to the curve

$$y^2 = 4\sqrt{2}x \text{ is } y = mx + \frac{\sqrt{2}}{m}$$

It is tangent to the circle $x^2 + y^2 = 1$

$$\therefore \left| \frac{\sqrt{2}/m}{\sqrt{1+m^2}} \right| = 1 \Rightarrow m = \pm 1$$

$$\therefore \text{ tangent are } y = x + \sqrt{2} \text{ and } y = -x - \sqrt{2}$$

Compare with $y = -ax + c$

$$\Rightarrow a = \pm 1 \text{ and } c = \pm \sqrt{2}$$

84. The sum $1 + \frac{1^3 + 2^3}{1 + 2} + \frac{1^3 + 2^3 + 3^3}{1 + 2 + 3} + \dots$
 $+ \frac{1^3 + 2^3 + 3^3 + \dots + 15^3}{1 + 2 + 3 + \dots + 15} - \frac{1}{2}(1 + 2 + 3 + \dots + 15)$ is equal to :

- A. 1240 B. 620
C. 660 D. 1860

Ans. B

Sol:

$$1 + \frac{1^3 + 2^3}{1 + 2} + \frac{1^3 + 2^3 + 3^3}{1 + 2 + 3} + \dots + \frac{1^3 + 2^3 + 3^3 + \dots + 15^3}{1 + 2 + 3 + \dots + 15} - \frac{1}{2}(1 + 2 + 3 + \dots + 15)$$

Let $S_1 - S_2$

$$S_1 = \sum_{n=1}^{15} \frac{1^3 + 2^3 + \dots + n^3}{1 + 2 + \dots + n} = \sum_{n=1}^{15} \frac{n(n+1)}{2}$$

$$S_2 = \frac{1}{2} \frac{15 \times 16}{2} = 16$$

Now, $S_1 - S_2$

$$= \sum_{n=1}^{15} \frac{n(n+1)}{2} - 60$$

$$= \frac{1}{2} \sum_{n=1}^{15} n^2 + \frac{1}{2} \sum_{n=1}^{15} n - 60$$

$$= \frac{1}{2} \times \frac{15 \times 16 \times 31}{6} + \frac{1}{2} \times \frac{15 \times 16}{2} - 60$$

$$= 620$$

85. The sum of the real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$,

is equal to:

A. -4

B. 0

C. 1

D. 6

Ans. B

Sol:

Given equation

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$$

After expanding,

$$x(-3x \times (x+2) - 2x(x-3)) + (-6)(2(x+2) + 3(x-3)) + (-1)(4x + 3(-3x))$$

$$\Rightarrow -5x^3 + 30x - 30 + 5x = 0$$

$$x^3 - 7x + 6 = 0$$

Since all roots are real

$$\therefore \text{Sum of roots} = -\frac{\text{coefficient of } x^2}{\text{coefficient of } x^3} = 0$$

$$\text{Sum of roots} = 0$$

86. If both the mean and the standard deviation of 50 observations $x_1, x_2, \dots, x_{50}$ are equal to 16, then the mean of $(x_1 - 4)^2, (x_2 - 4)^2, \dots, (x_{50} - 4)^2$, is :

- A. 525 B. 480
C. 400 D. 380

Ans. C

Sol:

It is given that both mean and standard deviation of 50 observations $x_1, x_2, x_3, \dots, x_{50}$ are equal to 16,

$$\text{Mean } (\mu) = \frac{\sum x_i}{50} = 16$$

$$\therefore \sum x_i = 16 \times 50$$

$$\text{and standard deviation } = (\sigma) = \sqrt{\frac{\sum x_i^2}{50} - (\mu)^2} = 16$$

$$\Rightarrow \frac{\sum x_i^2}{50} = 256 \times 2$$

$$\text{Required mean} = \frac{\sum (x_i - 4)^2}{50}$$

$$= \frac{\sum x_i^2 + 16 \times 50 - 8 \sum x_i}{50}$$

$$= 256 \times 2 + 16 - 8 \times 16$$

$$= 400$$

87. If $\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha$, where $-1 \leq x \leq 1, -2 \leq y \leq 2, x \leq \frac{y}{2}$, then for all $x, y, 4x^2 - 4xy \cos \alpha + y^2$ is equal to :

- A. $4 \sin^2 \alpha - 2x^2 y^2$
- B. $4 \cos^2 \alpha + 2x^2 y^2$
- C. $2 \sin^2 \alpha$
- D. $4 \sin^2 \alpha$

Ans. D

Sol: Given that

$$\cos^{-1} x - \cos^{-1} \frac{y}{2} = \alpha, \text{ where } -1 \leq x \leq 1,$$

$$-2 \leq y \leq 2 \text{ and } x \leq \frac{y}{2}$$

$$\therefore \cos^{-1} \left(x \frac{y}{2} + \sqrt{1-x^2} \sqrt{1-(y/2)^2} \right) = \alpha$$

$$\left[\because \cos^{-1} x - \cos^{-1} y = \cos^{-1} \left(xy + \sqrt{1-x^2} \sqrt{1-y^2} \right) \mid x \mid, \mid y \mid \leq \text{ and } x + y \geq 0 \right]$$

$$x \frac{y}{2} + \sqrt{1-x^2} \sqrt{1-\frac{y^2}{4}} = \cos \alpha$$

$$\left(\cos \alpha - \frac{xy}{2} \right) = \sqrt{1-x^2} \sqrt{1-\frac{y^2}{4}}$$

Squaring both sides

$$x^2 + \frac{y^2}{4} - xy \cos \alpha = 1 - \cos^2 \alpha$$

$$= \sin^2 \alpha$$

88. If z and w are two complex numbers such that $|zw| = 1$ and $\arg(z) - \arg(w) = \frac{\pi}{2}$, then :

- A. $z\bar{w} = \frac{-1+i}{\sqrt{2}}$
- B. $z\bar{w} = \frac{1-i}{\sqrt{2}}$
- C. $\bar{z}w = -i$
- D. $\bar{z}w = i$

Ans. C

Sol:

It is given that, there are two complex numbers z and w , such that $|zw| = 1$ and $\arg(z) - \arg(w) = \pi/2$

$$\therefore |z| |w| = 1 \quad [\because |z_1 - z_2| = |z_1| |z_2|]$$

$$\text{and } \arg(z) = \frac{\pi}{2} + \arg(w)$$

$$\text{Let } |z| = r, \text{ then } |w| = 1/r \dots (i)$$

$$|zw| = 1$$

$$|z| |w| = 1$$

$$\text{Let } w = \frac{1}{r} e^{i\theta}$$

$$\text{then } z = r e^{i\left(\theta + \frac{\pi}{2}\right)}$$

$$\bar{z}w = e^{-i\left(\theta + \frac{\pi}{2}\right)} \cdot e^{i\theta} = e^{-i(\pi/2)} = -i$$

$$\text{and } z\bar{w} = e^{-i\left(\theta + \frac{\pi}{2}\right)} \cdot e^{i\theta} = e^{-i\pi/2} = i$$

89. If the plane $2x - y + 2z + 3 = 0$ has the distances $\frac{1}{3}$ and $\frac{2}{3}$ units from the planes $4x - 2y + 4z + \lambda = 0$ and $2x - y + 2z + \mu = 0$ respectively, then the maximum value of $\lambda + \mu$ is equal to:

A. 13

B. 9

C. 5

D. 15

Ans. A

Sol:

Equation of given plane are

$$2x - y + 2z + 3 = 0 \dots (i)$$

$$4x - 2y + 4z + \lambda \dots (ii)$$

$$\text{and } 2x - y + 2z + \mu = 0 \dots (iii)$$

$\therefore$ Distance between two parallel planes

$$ax + by + cz + d_1 = 0, \text{ and}$$

$$ax + by + cz + d_2 = 0 \text{ is}$$

$$\text{distance} = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

Distance between planes (i) and (ii)

$$(i) \frac{|\lambda - 6|}{\sqrt{16 + 4 + 16}} = \left| \frac{\lambda - 6}{6} \right| = \frac{1}{3}$$

$$|\lambda - 6| = 2$$

$$\lambda = 8, 4$$

Distance between planes (i) and (iii)

$$(ii) \frac{|\mu - 3|}{\sqrt{4 + 4 + 1}} = \frac{2}{3}$$

$$|\mu - 3| = 2$$

$$\mu = 5, 1$$

$$\therefore (\mu + \lambda)_{\max} = 13$$

90. The locus of the centres of the circles, which touch the circle, $x^2 + y^2 = 1$ externally, also touch the y – axis and lie in the first quadrant, is:

A. $x = \sqrt{1 + 4y}, y \geq 0$

B. $y = \sqrt{1 + 4x}, x \geq 0$

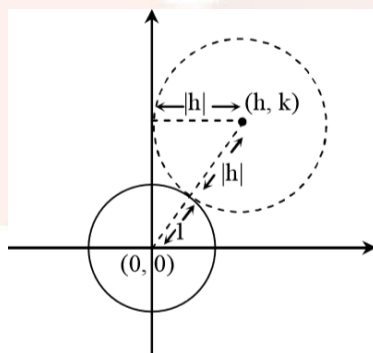
C. $x = \sqrt{1 + 2y}, y \geq 0$

D. $y = \sqrt{1 + 2x}, x \geq 0$

Ans. D

Sol:

Let (h, k) be the centre of the circle and radius $r = h$, as circle touch the Y . axis and other circle $x^2 + y^2 = 1$ whose centre $(0, 0)$ and radius is 1.



$$\sqrt{h^2 + k^2} = 1 + |h|$$

$h > 0$ and $k > 0$ for first quadrant.

$$h^2 + k^2 = 1 + h^2 + 2|h|$$

$$k^2 = 1 + 2|h|$$

$$y^2 = 1 + 2x$$

